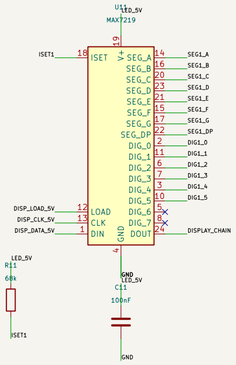
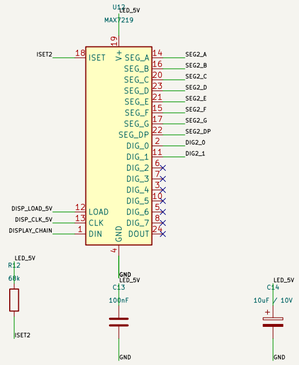
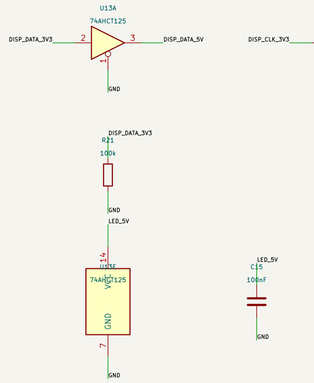
ARC-164 Power Driver Board: USB Teensy 4.1, display drivers, 5V distribution and backlight dimming

Teensy powered by USB. External regulated 5V feeds LEDs/MAX7219 only. VIN/VUSB are not joined to external 5V.





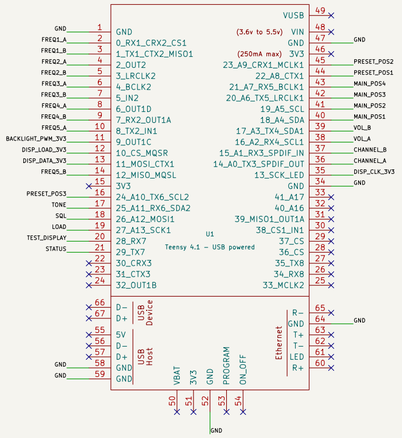


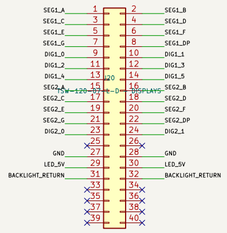
J20 cable goes directly to the unchanged LED Subboard. J10 goes to Main Board controls. J21 feeds Main Board backlights.

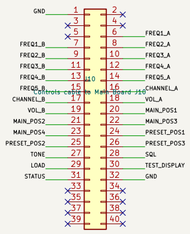
DCS-BIOS: MAX7219 intensity controls display brightness; GPIO9 PWM controls panel backlighting.

MAIN/PRESET use direct grounded contacts, no resistor ladders. Enable internal pull-ups for all 26 control GPIO inputs.

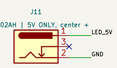
GPIO assignment is a new ARC-164 draft; firmware must be updated to match GPIO-Assignments.json.













19 green backlight LEDs and individual 330 ohm series resistors are on Main Board; physical clearance and brightness need prototype testing.

J20 DISPLAY ONLY: 40-pin 2.54mm IDC cable, 6in. Pins 1-32 retain existing assignments; 33-40 NC. Key toward PCB top edge. Label DISPLAY / J20; J10 is CONTROLS. Plain headers are not mechanically keyed.

Sheet: /
File: Power Driver Board.kicad_sch

Title: ARC-164 Power Driver Board — UNTESTED PROTOTYPE

Size: A2	Date:	**Rev: 1.0 Prototype**
KiCad E.D.A. 10.0.7		Id: 1/1